Time Series Analysis
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Assignment 4

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1 Parameter estimation in state-space model

1.1 Simulation of a One-Dimensional State-Space Model

We consider the state-space model

$$X_t = aX_{t-1} + b + e_t, \quad e_t \sim \mathcal{N}(0,1),$$

with parameters

$$a = 0.9, \quad b = 1, \quad X_0 = 5.$$

Theoretical Properties

This model corresponds to an autoregressive process of order one (AR(1)) with a constant drift term b. Since $|a| < 1$, the process is stable and admits a stationary distribution.
The stationary mean is given by

$$E[X_t] = \frac{b}{1-a} = \frac{1}{1-0.9} = 10.$$

The stationary variance is

$$\mathrm{Var}(X_t) = \frac{\sigma^2}{1-a^2} = \frac{1}{1-0.9^2}.$$

Simulation Description

Five independent realizations of the process are generated over $n = 100$ time steps. Each trajectory starts from the same initial value $X_0 = 5$ and evolves recursively according to the model equation.
The noise terms e_t are independent and identically distributed Gaussian random variables with mean 0 and variance 1.

Simulated Trajectories

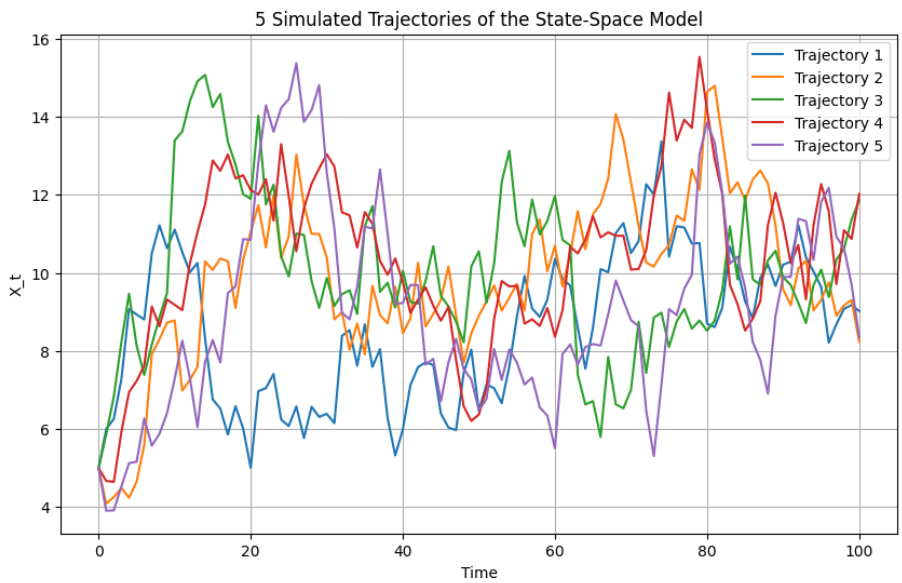

Figure 1: Five simulated trajectories of the state-space model.

Interpretation

Although all trajectories start from the same initial value, they diverge due to the stochastic noise. Because the system is stable ($|a| < 1$), all trajectories tend to fluctuate around the stationary mean value of 10 over time.

1.2 Simulation with Noisy Observations

We now consider the extended state-space model

$$X_t = aX_{t-1} + b + e_{1,t}, \quad e_{1,t} \sim \mathcal{N}(0, \sigma_1^2),$$

$$Y_t = X_t + e_{2,t}, \quad e_{2,t} \sim \mathcal{N}(0, \sigma_2^2),$$

with parameters $a = 0.9$, $b = 1$, $\sigma_1^2 = 1$, and $\sigma_2^2 = 1$.

Description of the Simulation

A single realization of the latent state process $\{X_t\}$ is generated recursively starting from $X_0 = 5$. At each time step, an observation Y_t is obtained by adding Gaussian noise to the true state X_t.
Thus, while X_t represents the true (hidden) system state, Y_t corresponds to a noisy measurement of that state.

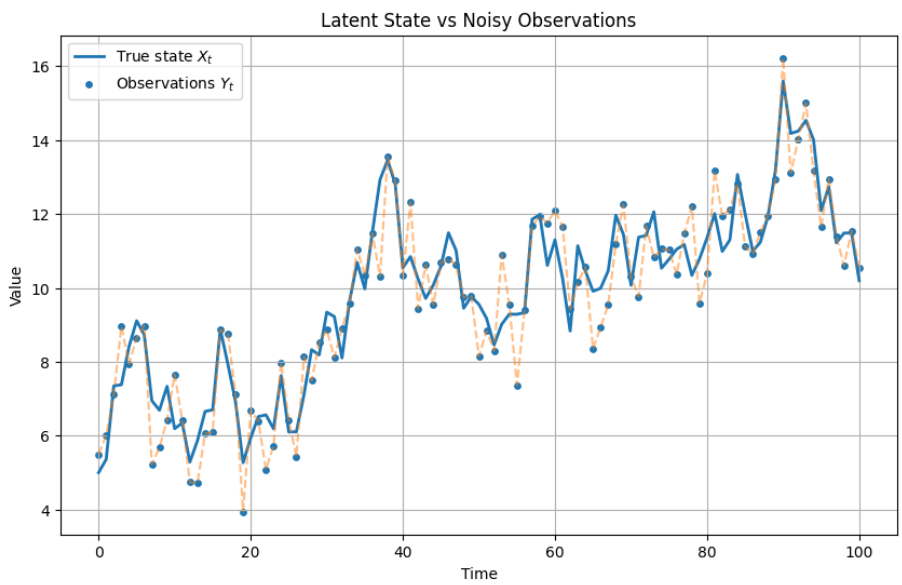

Figure 2: Simulated latent state trajectory Xt and the observations Yt

Discussion

The plot illustrates that the observations Y_t follow the general trend of the true state X_t, but are corrupted by noise. In particular, the observations fluctuate more strongly and do not lie exactly on the true trajectory.
Although the underlying process is smooth due to the autoregressive structure, the measurement noise introduces additional variability, making the observed data appear more irregular.
This demonstrates a key challenge in practice: the true state of a system is not directly observable, and inference must rely on noisy measurements. As a result, standard estimation techniques such as ordinary least squares are not directly applicable.
Instead, methods like the Kalman filter are required to optimally combine the model dynamics with the noisy observations in order to estimate the hidden state and unknown parameters.

1.3 Parameter Estimation and Kalman Filtering

Implementation and Methodology

The Kalman filter for the 1D state-space model was implemented using a recursive Predict-Update cycle. Given the scalar parameters $a = 0.9, b = 1, \sigma_1^2 = 1$ and $R = 1$, the filter computes the following steps for each time step t:

- **Prediction Step:** $\hat{X}_{t|t-1} = a\hat{X}_{t-1|t-1} + b$ and $\hat{\Sigma}^{xx}_{t|t-1} = a^2\hat{\Sigma}^{xx}_{t-1|t-1} + \sigma_1^2$.
- **Innovation:** $\nu_t = Y_t - \hat{X}_{t|t-1}$ with variance $\hat{\Sigma}^{yy}_{t|t-1} = \hat{\Sigma}^{xx}_{t|t-1} + R$.
- **Update Step:** $\hat{X}_{t|t} = \hat{X}_{t|t-1} + K_t\nu_t$ where $K_t = \hat{\Sigma}^{xx}_{t|t-1}/\hat{\Sigma}^{yy}_{t|t-1}$.

Results and Discussion

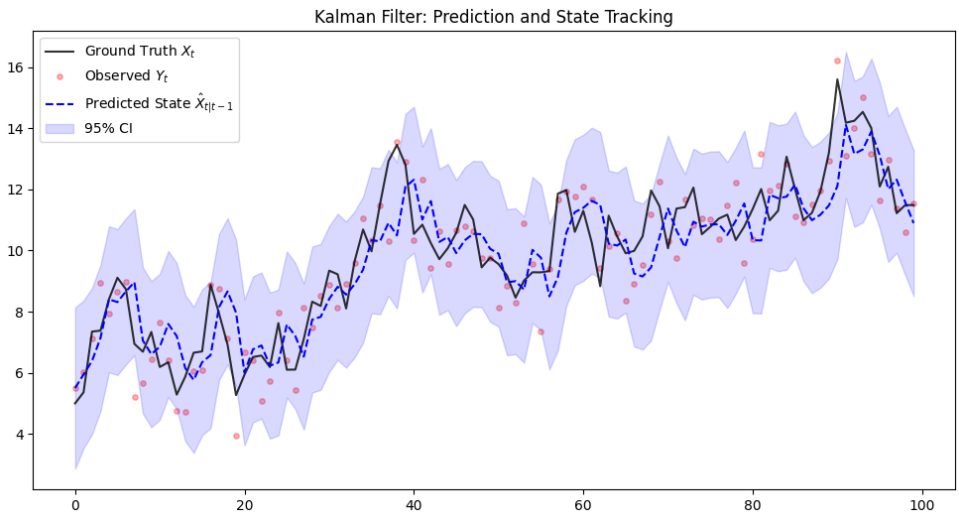

Figure 3: True state X_t, noisy observations Y_t, and predicted state $\hat{X}_{t|t-1}$ with 95% confidence interval.

The visualization shows that the predicted state $\hat{X}_{t|t-1}$ provides a much smoother estimate than the raw observations Y_t.

- **Tracking Performance:** Despite the significant measurement noise ($\sigma_2^2 = 1$), the filter successfully tracks the latent state X_t. The predicted mean follows the general trend of the ground truth without being overly sensitive to individual outliers in Y_t.
- **Predicted Distributions:** The 95% confidence interval, calculated as $\hat{X}_{t|t-1} \pm 1.96\sqrt{\hat{\Sigma}^{xx}_{t|t-1}}$, consistently contains the true latent state. This indicates that the predicted distributions are well-calibrated to the simulated data.
- **Steady State:** It is observed that the prediction variance $\hat{\Sigma}^{xx}_{t|t-1}$ converges quickly to a steady-state value, reflecting the balance between the uncertainty added by the process noise and the information gained from measurements.

1.4 Parameter Estimation using Maximum Likelihood

In this section, we implement the Maximum Likelihood Estimation (MLE) framework to recover the parameters $\theta = (a, b, \sigma_1)$ from noisy observations. The negative log-likelihood is computed using the innovations ϵ_t and their variances S_t obtained from the Kalman Filter:

$$-\ln L(\theta) = \frac{1}{2} \sum_{t=1}^{n} \left[\ln(2\pi S_t) + \frac{\epsilon_t^2}{S_t} \right]$$

By minimizing this function using the `L-BFGS-B` algorithm, we obtain the parameter estimates for 100 independent realizations.

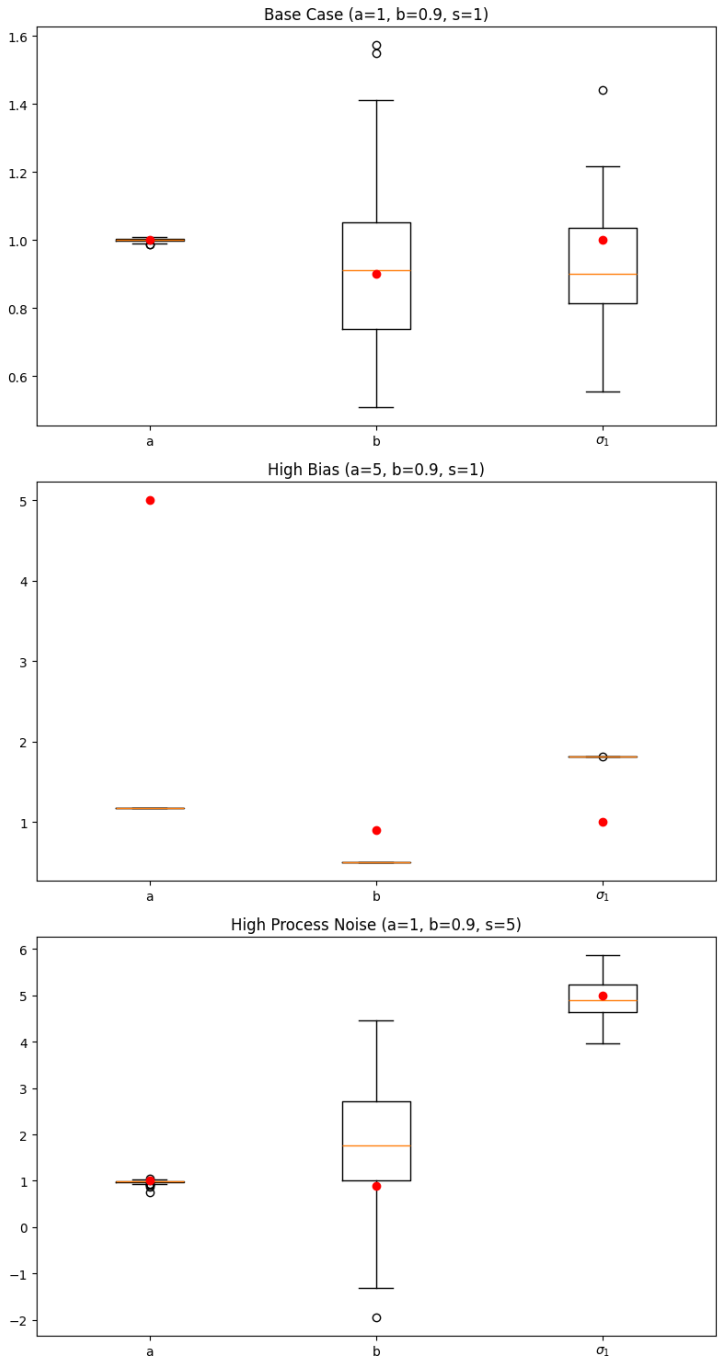

Figure 4: Parameter estimates across 100 realizations for the three specified cases. Red dots indicate the true parameter values used for simulation.

Comparison Across Cases

- **Base Case ($a = 1, b = 0.9, \sigma_1 = 1$):** The estimates align well with the true values. The parameter a is estimated with very high precision, while b and σ_1 show a larger spread, indicating higher uncertainty in their estimation.
- **High Bias Case ($a = 5, b = 0.9, \sigma_1 = 1$):** This case demonstrates the limits of the estimation framework. Since $a = 5$ exceeds the stability threshold ($|a| < 1$), the system state grows exponentially. The optimizer fails to recover the true values, leading to a significant bias in all estimated parameters. This highlights the importance of model stability for reliable parameter recovery.
- **High Process Noise Case ($a = 1, b = 0.9, \sigma_1 = 5$):** With increased process noise, the variance of the estimates for b and σ_1 increases substantially. While the median of σ_1 remains close to the truth, the estimation of the intercept b becomes very unreliable due to the heavy fluctuations in the state.

Estimation Difficulty and Alignment

The results show that the parameters do not align equally well across all cases:

- **Hardest Parameter:** The bias b is consistently the most difficult parameter to estimate, showing the widest interquartile ranges. This is due to its sensitivity to noise and its coupling with the transition parameter a.
- **Alignment:** While the Gaussian MLE is effective for stable systems (Base Case), it becomes highly inaccurate in unstable regimes ($a = 5$).
- **Sensitivity:** Parameter a is generally estimated with the highest precision in stable or marginally stable ($a = 1$) systems, as the temporal correlation provides a very distinct signature in the data.

1.5 Robustness Analysis: Heavy-Tailed System Noise

In this section, we investigate the robustness of the Kalman filter and Maximum Likelihood Estimation (MLE) framework when the fundamental assumption of Gaussian noise is violated. We simulate the system using a Student's t-distribution for the process noise while maintaining the Gaussian assumption in the estimation pipeline.

Theoretical Comparison: Normal vs. Student's t-distribution

The Student's t-distribution is defined by the degrees of freedom parameter ν. As $\nu \to \infty$, the distribution converges to the standard Normal distribution. Figure 5 illustrates the density functions for various values of ν.

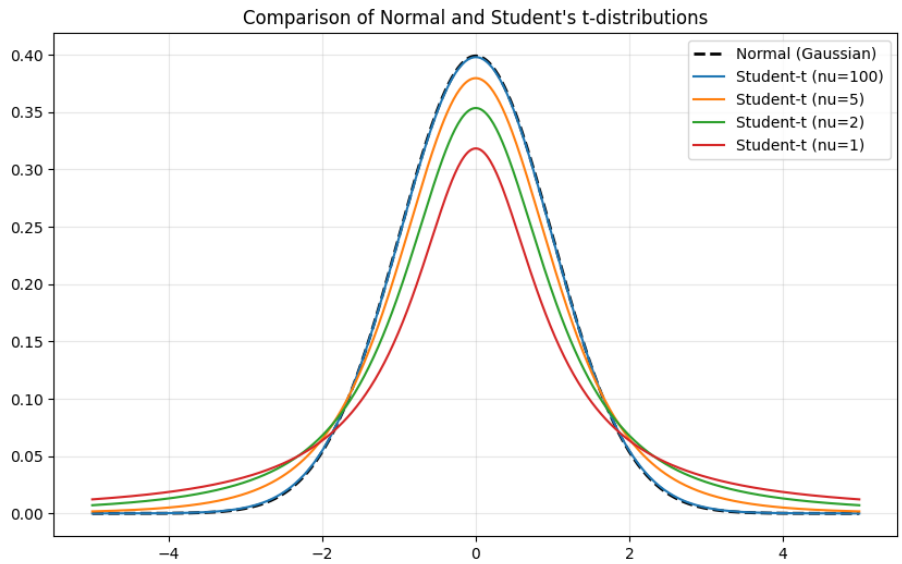

Figure 5: Comparison of the standard Normal distribution and Student's t-distributions with different degrees of freedom (ν).

As seen in Figure 5, as ν decreases, the distribution exhibits heavy tails. This implies a significantly higher probability of extreme events compared to the Gaussian distribution. In a Kalman filtering context, these large shocks are problematic because the filter, optimized for Gaussian noise, tends to overreact to outliers. It misinterprets these t-shocks as significant structural shifts in the state X_t, which distorts the likelihood and leads to biased parameter estimates.

Simulation Results and Impact on Estimation

We conducted 100 realizations for $n = 100$ time steps across $\nu \in \{100, 5, 2, 1\}$. The results are summarized in Figure 6.

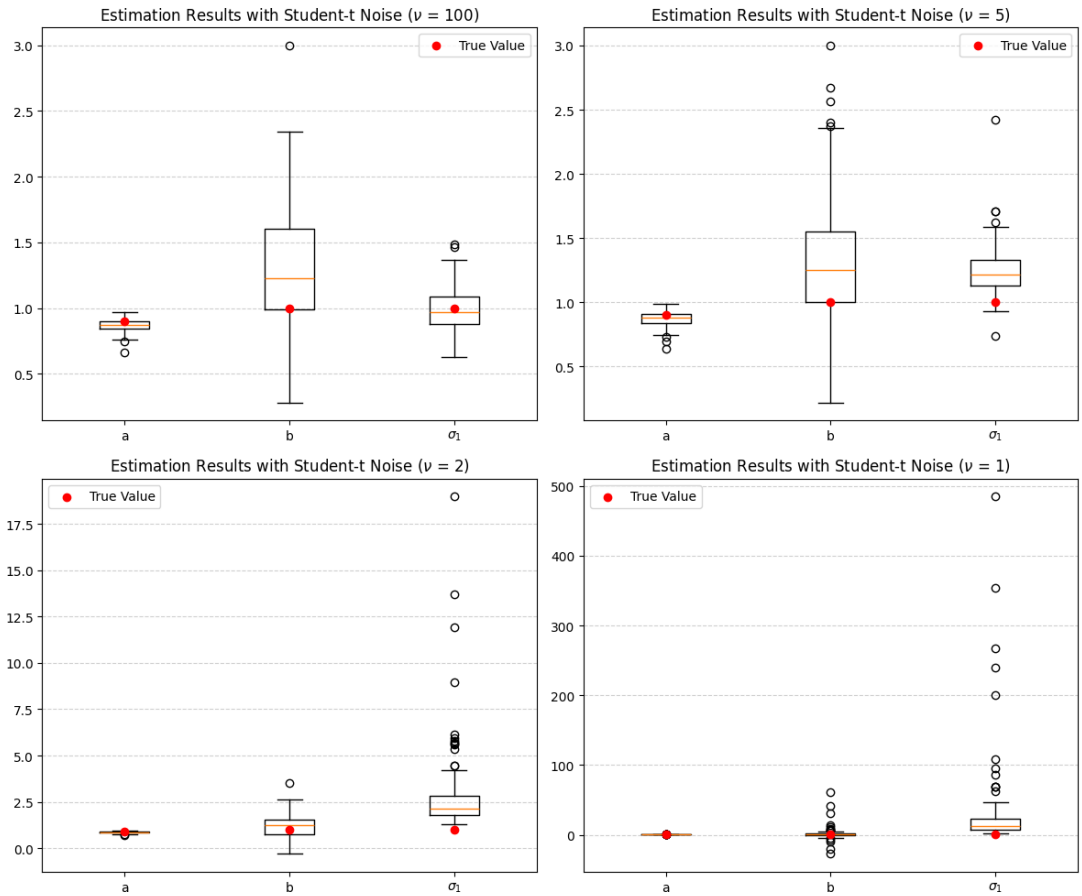

Figure 6: Parameter estimates under Student's t-distributed process noise for different degrees of freedom ν. Red dots represent true values.

Discussion

The boxplots reveal that as ν decreases, the precision and accuracy of the MLE framework degrade:

- **Variance Overestimation:** The estimates for σ_1 increase dramatically as ν drops. This is because the Gaussian-based model tries to capture the heavy tails of the t-distribution by inflating the overall variance σ^2 of the bell curve to cover the observed outliers.
- **Instability in b:** The intercept b shows extreme volatility. Individual extreme shocks pull the estimated trajectory away from the truth, forcing the optimizer to choose extreme values for b to maximize the fit.

For $\nu > 2$, the variance of the t-distribution is $\mathrm{Var}(\lambda) = \frac{\nu}{\nu-2}$. As $\nu \to 2$, the theoretical variance approaches infinity. For $\nu = 1$ (Cauchy distribution), neither the mean nor the variance is defined. This explains the extreme values observed in the σ_1 boxplot for $\nu = 1$, where estimates reach magnitudes of several hundreds.

Conclusion

The study demonstrates that the standard Kalman filter is **not robust** to non-Gaussian, heavy-tailed noise. In practical applications like finance or sensor data processing where outliers are common, relying on Gaussian assumptions can lead to highly unreliable estimates and an overestimation of system risks.

2 Parameter estimation in state-space model

2.1 Exploratory Analysis of the Data

The dataset contains four time-dependent variables: transformer temperature Y_t, outdoor temperature $T_{a,t}$, solar radiation S_t, and electrical load I_t. In this section, we perform an initial visual inspection to identify patterns, trends, and potential relationships between the variables.

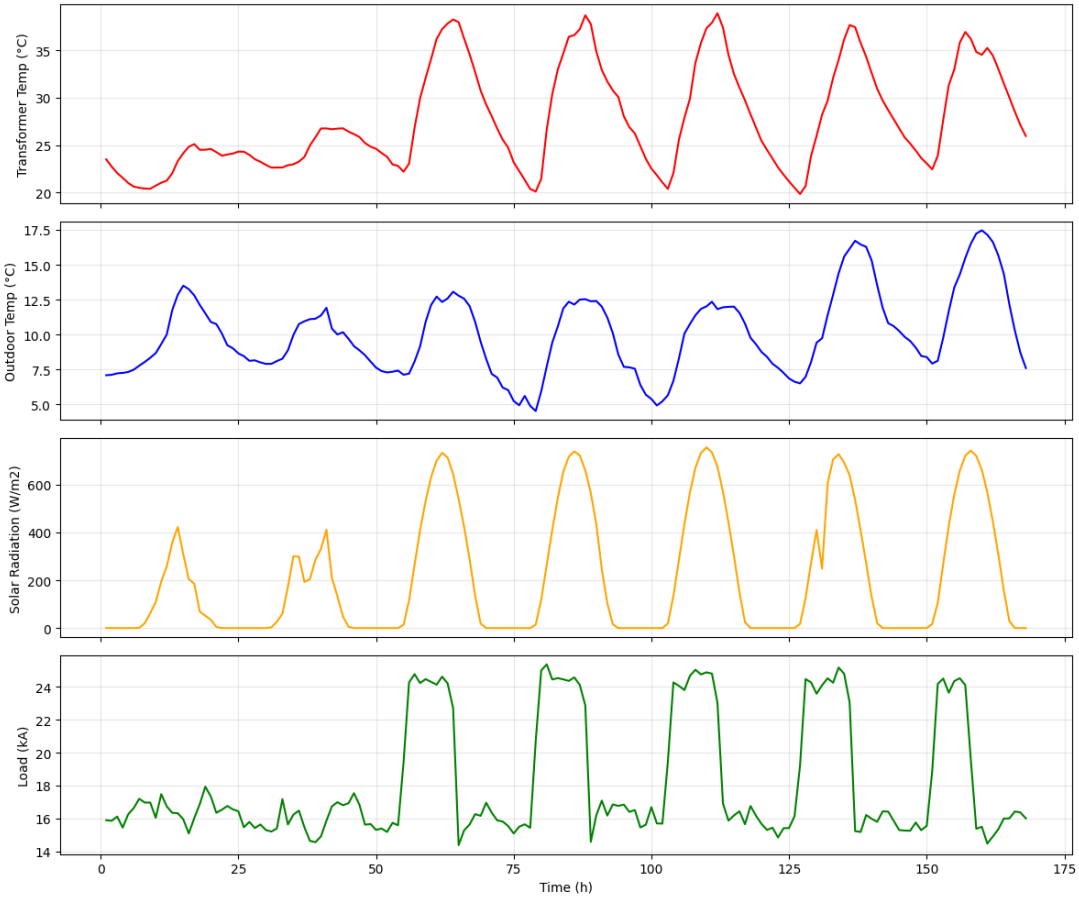

Figure 1: Transformer Data

Transformer Temperature

The transformer temperature Y_t exhibits smooth and cyclical behavior over time, ranging approximately between 20°C and 38°C. The dynamics are clearly not random but instead show a structured temporal pattern with gradual increases and decreases. This indicates a strong thermal inertia in the system, meaning that temperature changes occur gradually rather than instantaneously.

Outdoor Temperature

The outdoor temperature $T_{a,t}$ shows a pronounced daily periodic pattern, varying between approximately 5°C and 17°C. The fluctuations are smooth and regular, consistent with a diurnal cycle. A visual comparison with Y_t suggests a positive relationship, although the transformer temperature responds more slowly to changes in ambient conditions.

Solar Radiation

The solar radiation S_t follows a highly regular day-night cycle. It remains close to zero during nighttime and exhibits sharp peaks during daylight hours. This variable is strongly exogenous and deterministic in structure, reflecting environmental energy input into the system.

Electrical Load

The electrical load I_t shows a more irregular and step-like behavior compared to the other variables. It alternates between low and high regimes, suggesting changes in operational demand over time. Periods of high load appear to coincide with increases in transformer temperature, indicating a strong potential influence on thermal behavior.

Discussion of Relationships

From the visual inspection, several relationships can be identified:

- The strongest apparent driver of transformer temperature Y_t is the electrical load I_t.
- Outdoor temperature $T_{a,t}$ also appears positively correlated with Y_t, but with a smoother and more delayed effect.
- Solar radiation S_t influences the system indirectly through environmental heating effects.

Overall, the system exhibits clear multivariate dependencies and time-lagged interactions. This motivates the use of a state-space modeling framework to properly capture the underlying latent dynamics.

2.2 Residual Analysis, Parameter Interpretation, and Model Diagnostics

The transformer temperature dynamics were modelled using a one-dimensional discrete-time state-space model. The estimated system is defined as

$$x_{t+1} = Ax_t + Bu_t + Ge_{1,t} \tag{2.1}$$

$$y_t = Cx_t + e_{2,t}, \tag{2.2}$$

where:

- x_t denotes the latent thermal state of the transformer at time t,
- y_t is the measured transformer temperature,
- u_t is the vector of exogenous inputs,
- $e_{1,t} \sim \mathcal{N}(0, 1)$ is the process disturbance,
- $e_{2,t} \sim \mathcal{N}(0, \sigma_y^2)$ is the measurement noise.

The input vector is defined as

$$u_t = \begin{bmatrix} T_{a,t} \\ S_t \\ I_t \end{bmatrix}, \tag{2.3}$$

where:

- $T_{a,t}$ = ambient temperature,
- S_t = solar radiation,
- I_t = transformer load current.

The input matrix is estimated as

$$B = \begin{bmatrix} b_1 & b_2 & b_3 \end{bmatrix}. \tag{2.4}$$

Hence, the state equation can be written explicitly as

$$x_{t+1} = Ax_t + b_1 T_{a,t} + b_2 S_t + b_3 I_t + Gw_t. \tag{2.5}$$

Parameter Estimates and Numerical Interpretation

The model parameters were estimated by maximizing the log-likelihood function using the L-BFGS-B optimization algorithm. The convergence was successful, yielding the following estimates:

Parameter	**Symbol**	**Estimated Value**
System Matrix	A	0.8010
Ambient Temp. Coeff.	b_1	0.1011
Solar Radiation Coeff.	b_2	0.0028
Load Current Coeff.	b_3	0.2115
System Noise Scaling	G	0.6385
Output Matrix	C	1.0092
Measurement Noise Std.	σ_y	2.21×10^{-5}
Initial State	x_0	23.56

Table 2.1: Maximum Likelihood Estimates for the 1D State-Space Model.

The estimated system matrix $A \approx 0.8$ suggests a moderate thermal memory, where approximately 80% of the previous state is retained each hour. The input coefficients in B confirm physical expectations:

the transformer load ($b_3 \approx 0.21$) is the most significant driver of temperature increase, followed by the ambient temperature ($b_1 \approx 0.10$). The solar radiation ($b_2 \approx 0.003$) has a negligible direct effect.
Notably, the measurement noise σ_y is estimated to be near zero. This indicates that the Kalman Filter attributes almost all deviations to the process noise G, effectively treating the temperature measurements as highly accurate while allowing the state x_t to adapt flexibly to the observed data.

Model Performance and Visualization

The primary objective was to predict the temperature evolution accurately. Figure 2 illustrates the comparison between the observed data and the model's filtered predictions.

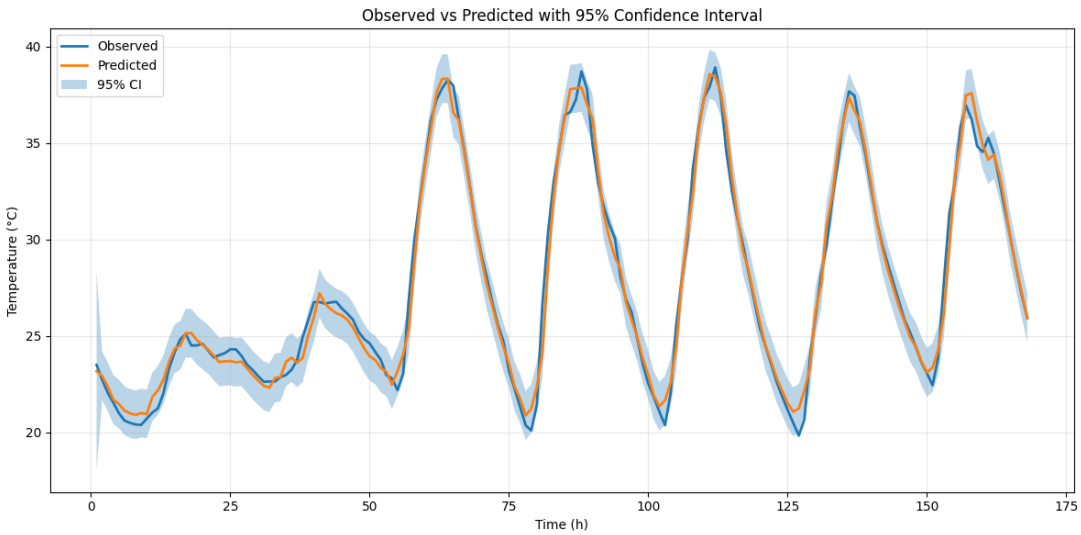

Figure 2: Observed vs. Predicted Temperature: The model shows high fidelity in tracking both diurnal cycles and load-induced peaks.

The visual fit is excellent, with the predicted trajectory closely mimicking the measured temperature. This suggests that even a 1-dimensional model is sufficient for general trend tracking and operational monitoring.

Residual Analysis and Diagnostics

To assess the statistical validity of the model, we analyze the innovations (residuals). Standard diagnostics are presented in Figure 3.

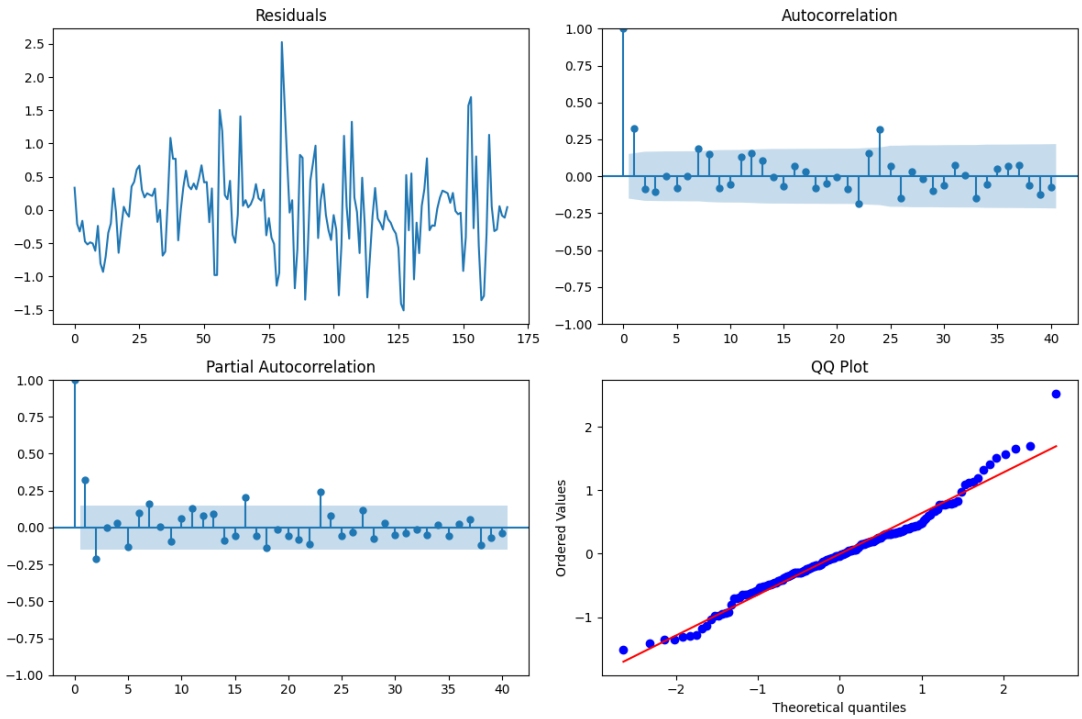

Figure 3: Residual Diagnostics: (Top-Left) Residual time series, (Top-Right) ACF, (Bottom-Left) PACF, and (Bottom-Right) Normal QQ-Plot.

The following observations are made from the diagnostics:

1. **Autocorrelation:** The ACF and PACF plots show a significant spike at lag 1. This indicates that the residuals are not pure white noise, implying that some temporal dynamics (likely higher-order thermal effects) are not fully captured by the 1D state-space structure.
2. **Normality:** The QQ-plot shows that the residuals are approximately normal in the center but deviate at the tails.
3. **Stationarity:** The residual time series appears to fluctuate around zero, though some clusters of higher variance are visible during periods of rapid temperature change.

Model Selection Criteria

To provide a baseline for future model comparisons, the information criteria were calculated:

- **Log-Likelihood:** -165.50
- **AIC:** 347.00
- **BIC:** 371.99

The relatively low AIC/BIC values, combined with the visual fit, suggest that the model is well-parameterized for a first-order approximation.

Conclusion

The 1D state-space model captures the transformer temperature dynamics well and shows strong agreement between predicted and observed values. Load current is identified as the dominant influence, while ambient temperature has a moderate effect and solar radiation is negligible. Residual analysis indicates remaining autocorrelation, suggesting that some system dynamics are not fully captured. Overall, the model provides a solid first-order approximation but could be improved with a more complex state representation.

2.3 2D State-Space Expansion

To account for the uncaptured dynamics observed in the 1D model's residuals—specifically the significant autocorrelation at the first lag—the system was expanded to a two-dimensional latent state space $X_t \in R^2$. This allows the model to differentiate between distinct thermal processes, such as the rapid heating of the cooling oil versus the slower thermal inertia of the transformer's massive core.

Model Formulation

The generalized system equations for the 2D model are defined as:

$$X_{t+1} = AX_t + Bu_t + G\mathbf{e}_{1t} \tag{2.6}$$

$$Y_t = CX_t + e_{2t} \tag{2.7}$$

where $A \in R^{2\times 2}$ represents the state transition matrix, $B \in R^{2\times 3}$ relates the exogenous inputs (ambient temperature, solar radiation, and load current) to the state evolution, and $C \in R^{1\times 2}$ maps the internal states to the observed temperature.

Parameter Estimation and Optimization

The parameters were estimated using Maximum Likelihood Estimation (MLE). The estimates are summarized below:

- **Transition Matrix (A):** $\begin{bmatrix} 2.817 & 3.066 \\ -1.889 & -2.049 \end{bmatrix}$
- **Input Matrix (B):** $\begin{bmatrix} -0.177 & -0.003 & -0.897 \\ 0.276 & 0.005 & 1.096 \end{bmatrix}$
- **Output Matrix (C):** $\begin{bmatrix} 0.798 & 0.712 \end{bmatrix}$
- **Noise Parameters:** $G \approx 0.472$, $\sigma_y \approx 0.0082$

The eigenvalues of A determine the system stability and response time. The interaction between the two states suggests a coupled system where one state acts as a buffer, capturing the delayed thermal response that the 1D model previously attributed to process noise.

Performance and Residual Diagnostics

The 2D model's predictive performance and residual characteristics are illustrated in Figures 4 and 5.

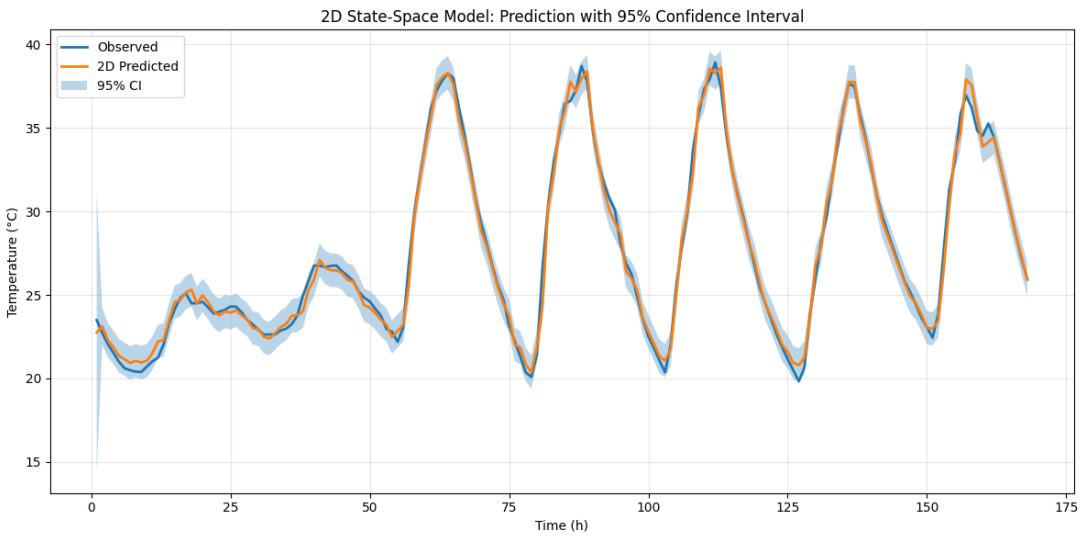

Figure 4: 2D State-Space Model Fit: The multi-dimensional state allows for a more nuanced tracking of rapid temperature fluctuations and thermal inertia.

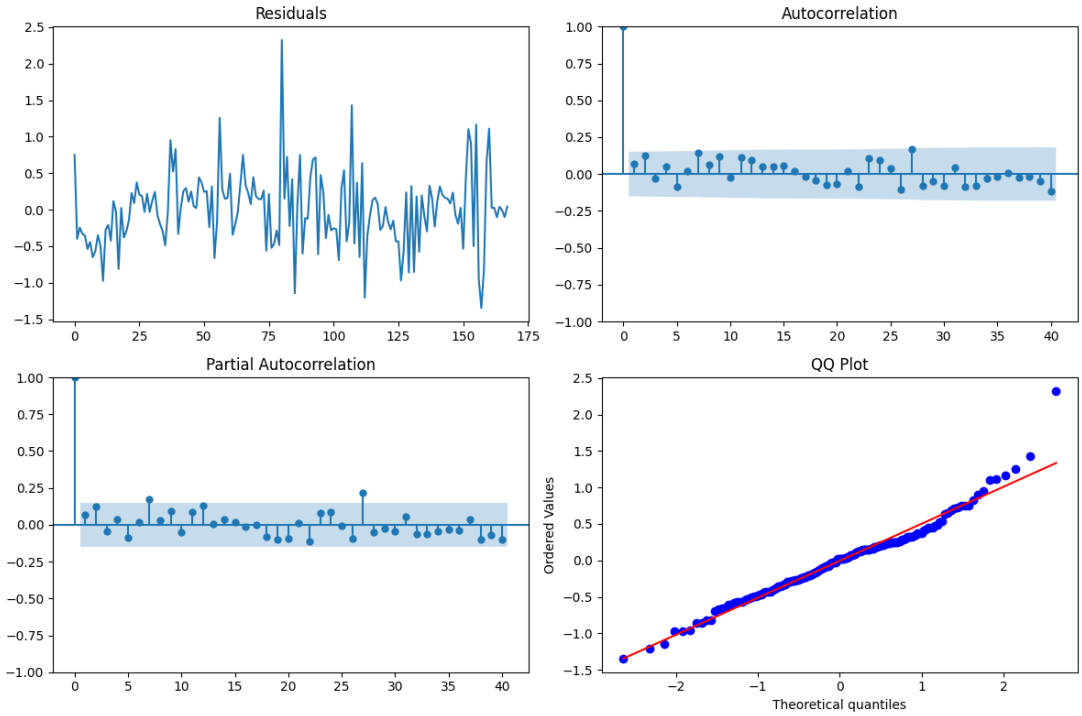

Figure 5: 2D Residual Diagnostics: The autocorrelation is significantly reduced compared to the 1D model.

Comparative Discussion

Comparing the 2D model to the initial 1D approach yields the following insights:

1. **Tracking of Spikes:** The 1-step predictions of the 2D model track temperature spikes with higher precision. The dual-state structure effectively captures the overshoot and cooling curves during sudden changes in load current (I_t).

2. **Residual Autocorrelation:** A critical improvement is observed in the ACF and PACF plots. The significant lag-1 autocorrelation present in the 1D model has been substantially mitigated, indicating that the second state successfully internalizes the previously unexplained temporal dependencies.

3. **Information Criteria:**

   - **Log-Likelihood:** -126.28
   - **AIC:** 284.55
   - **BIC:** 334.54

   The reduction in both AIC and BIC confirms that the added complexity of the 2D model is statistically justified by the gain in explanatory power.

Physical Interpretation of Latent States

The two latent states can be interpreted through a physical lens. One state likely represents the Top Oil Temperature, which reacts quickly to changes in ambient conditions and load. The second state likely represents the Winding or Core Temperature, characterized by a much larger thermal mass. This state acts as a thermal buffer, storing energy and releasing it slowly, which explains why the 2D model handles the transition between heating and cooling phases more gracefully than the 1D model.

Conclusion

The transition to a 2D state-space model represents a significant enhancement in modeling transformer thermal dynamics. By capturing the interaction between different thermal components, the model reduces residual autocorrelation and provides a more robust framework for real-time temperature monitoring and predictive maintenance.

2.4 Interpretation of Reconstructed Latent States

Based on the Kalman filter estimation, we now investigate the physical meaning of the two-dimensional latent state vector

$$\mathbf{x}_t = [x_{1,t}, x_{2,t}]^T.$$

The interpretation is based on the reconstructed state trajectories as well as the estimated system matrices.

State Trajectories and Input Influence

Figure 6 shows the reconstructed latent states over time.

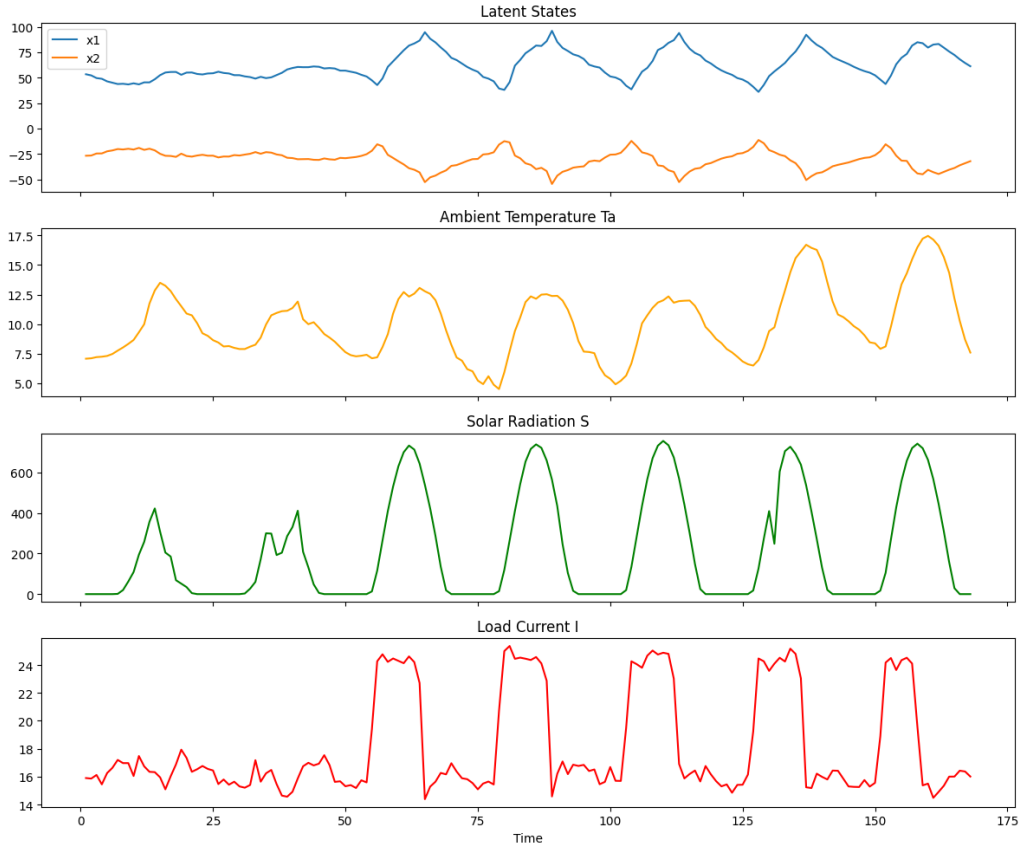

Figure 6: Reconstructed latent state trajectories (x_1 and x_2).

The trajectories indicate that both states respond strongly to changes in the load current I. During the first 50 time steps, the system remains relatively stable. Once periodic load pulses begin around $t \approx 55$, both states start to oscillate significantly.
A notable feature is the clear inverse relationship between the two states: when x_1 increases, x_2 decreases, and vice versa. This suggests that the states capture two coupled but opposing thermal mechanisms within the system.

Interpretation of the Estimated Matrices

The estimated transition matrix is

$$A = \begin{bmatrix} 2.817 & 3.066 \\ -1.889 & -2.049 \end{bmatrix}.$$

The relatively large off-diagonal elements indicate strong interaction between the two states. This means that each state strongly influences the future evolution of the other. Such coupling is plausible in thermal systems where internal heat generation and external cooling processes are closely linked.
The estimated input matrix is

$$B = \begin{bmatrix} -0.177 & -0.003 & -0.897 \\ 0.276 & 0.005 & 1.096 \end{bmatrix},$$

where the columns correspond to ambient temperature T_a, solar radiation S, and load current I.
The largest coefficients appear in the third column, showing that the load current is the dominant external driver of the system. This is physically reasonable, since transformer heating is mainly caused by electrical losses under load. In contrast, the coefficients related to solar radiation are close to zero, indicating only minor influence.

Physical Meaning of the Two States

Based on the observed dynamics, the two latent states can be interpreted as follows:

State x_1: Internal/Core Temperature The first state likely represents the internal transformer temperature, such as the winding or core temperature. It exhibits large variations during load pulses and reflects the main heat accumulation process.

State x_2: Cooling or Buffer State The second state likely captures the thermal buffer or cooling mechanism, for example the oil temperature, casing temperature, or heat-transfer gradient to the surroundings. Its inverse movement relative to x_1 suggests that it responds to internal heating by increasing heat dissipation.

Overall Assessment

The estimated model appears physically meaningful. The load current is correctly identified as the most important input, while solar radiation has only limited impact. The strong coupling between the states is consistent with a system in which internal heat generation and external cooling continuously interact.
Overall, the two-state model provides a realistic representation of transformer thermal dynamics by separating slow heat storage effects from faster cooling or buffer processes.